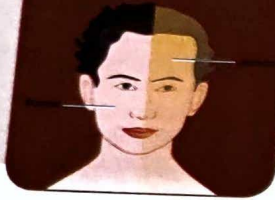


CHAPTER

26



Kamala

Learning Objectives

- ⇒ To learn the concept of Kamala
- ⇒ To learn the Nidana and Samprapti of Kamala in Ayurveda literatures
- ⇒ To frame out the probable Samprapti Ghataka based on the progression of the disease
- ⇒ To make the classification of Kamala based on Vidhi Samprapti
- ⇒ To learn the differences between Koshthashakhashrita and Shakhashrita Kamala
- ⇒ To enlist the Upadrava of Kamala
- ⇒ To learn the differences between Svatantra, Paratantra; and Alpapitta and Bahupitta Kamala

Introduction

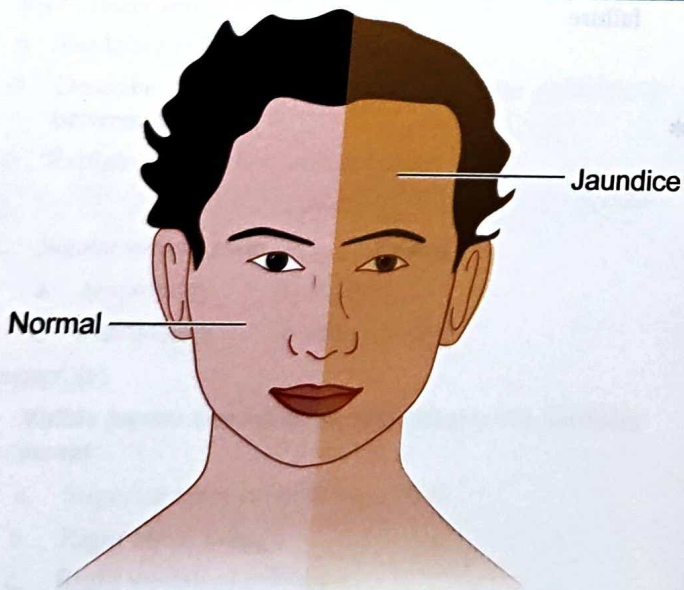


Figure 72: Image of patient of normal and jaundiced individuals

Kamala is *Pitta* dominant disease where *Rakta Dhatu Kshaya* (decrease of *Rakta Dhatu*) is seen. As per *Charaka*, it is abnormal *Rakta* responsible for the manifestation of *Kamala*. This is associated with yellowish discolouration of different body parts. *Kamala* is said to be leading to the involvement of *Raktavaha Srotas*, as *Yakrit* and *Pliha* are considered to be the *Srotomula* of *Raktavaha Srotas*. *Charaka* says two types of *Kamala* – *Shakhashrita Kamala* and *Koshthashakhashrita Kamala*. *Kumbhakamala*, *Halimaka*, *Panaki* are different stages of *Kamala* itself.

Nidana and Samprapti of Koshtha Shakhashrita Kamala

पाण्डुरोगी तु योऽत्यर्थं पित्तलानि निषेवते।
तस्य पित्तमसृग्मांसं दग्ध्वा रोगाय कल्पते॥३४॥
हारिद्रनेत्रः स भृशं हारिद्रत्वङ्नखाननः।
रक्तपीतशकृन्मूत्र मेकवर्णो हतेन्द्रियः॥३५॥
दाहाविपाकदौर्बल्य सदनारुचिकर्षितः।

कामला बहुपित्तैषा कोष्ठशाखाश्रया मता॥३६॥ Ca.Ci. – 16/34-36

- ⊛ A person suffering from disease *Pandu* is continuously consuming *Pitta* vitiating diet¹

- 1 When chyme is entering the small intestine, they move further to the distal part of intestine at a velocity of 0.5 to 2 cm/ second. The movement is faster at proximal intestine and slower at the distal intestine. Normally, peristaltic movements die out after travelling 3 to 5 cms, very rarely farther than 10 cm, so that forward movement of chyme is very slow which averages about 1cm/ second.
 - When the chyme enters the small intestine, that will be responsible for the secretion of cholecystokinin from mucosal cells of small intestine and this is stimulating the secretions of liver and pancreas.
 - Liver cells synthesise about 6 gms of bile salts daily. Bile salts reach duodenum along with biliary secretion. About 94% of the bile salts

• Pitta vitiation occurs inside the person, this vitiated Pitta will burn the Rakta (blood) and Mamsa (flesh)² and manifests Kamala.

• This is associated with yellowish discolouration of eyes, skin, nails and mouth³.

• This may also be associated with reddish yellow discolouration of urine, faeces.⁴

• The skin may resemble the colour of frog of rainy season⁵.



Figure 73: The colour of Bheka Varna (colour of frog of rainy season)

• The all Indriya of a suffering individual will lose their function⁶.

are reabsorbed into blood from small intestine. About 47% of the bile salts absorbed through early portions of small intestine and another 47% of the bile salts are absorbed through ileum. These salts enter portal blood and reenter the liver and these bile salts enter again to the bile to be secreted into the duodenum. Thus, 94% of the bile salts undergo 17 times of enterohepatic circulation before entering into the faecal matter.

- When an individual consumes Tikshna (highly irritating substances such as green chilli or red chilli), this is responsible for the inflammation of the stomach, small intestine and increased peristaltic activity in the alimentary canal. This is responsible for the much rapid movement of substances in the small intestine. This increases the secretion of bile more than 1000 ml/ day due to the influence of cholecystokinin. Due to the increased peristaltic movements, more amount of bile will be excreted in the faecal matter leading to the more yellowish discolouration of faecal matter.

2 While speaking about the pathogenesis of the Kamala, Rakta and Mamsa Dhatu affliction are observed. Here, the term Rakta Dhatu may be indicative of the destruction of blood cells resulting in the increased bilirubin and biliverdin formation, where as Mamsa Dhatu can be attributable to the involvement of gall bladder during the manifestation of the disease, not Mamsa Dhatu of all over the body.

3 As far the progression of the disease, appearance of jaundice first seen in the blood, later it can be seen in urine, later the discolouration seen in tongue and conjunctiva and then in the skin uniformly.

4 When there is free haemoglobin available in the circulating blood which may occur along with destruction of senescent RBCs, it can be transferred into the nephron easily leading to the reddish appearance of urine. In this occasion, yellowish appearance of urine sample indicates the presence of conjugated bilirubin in the urine.

Reddish appearance of faecal matter may indicate the bleeding occurring in the lower alimentary canal and as it is occurring in the lower alimentary canal, mixture of hydrochloric acid does not occur because of which there is no formation of acid haematin leading to dark tarry stool, the person may have reddish appearance of faecal matter. Yellowish appearance of faecal matter may indicate the presence of either inflammation of lower alimentary canal which leads to faster peristalsis or indicates hepatocellular jaundice or features of partial obstructive jaundice. In case of complete obstructive jaundice, the person may have whitish appearance of faecal matter due to the absence of bilirubin in the faecal matter.

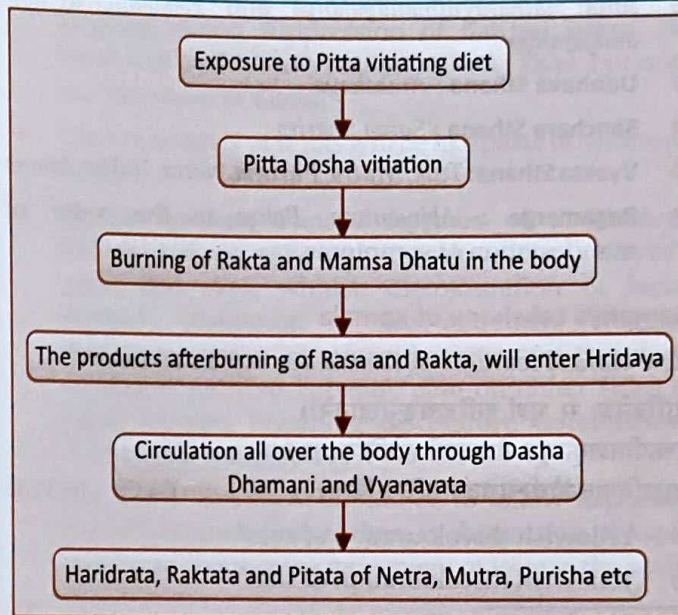
5 Indriya will function in a proper manner when they receive adequate amount of oxygen via RBCs. In an individual suffering from Kamala, the main pathological event is destruction of RBCs (either senescent or mature or immature RBCs depending upon the pathological process), thus one of the major function of RBC i.e. nourishing the Indriya (sense organs and organs of action) are hampered. Thus, he will have loss of functions of Indriyas.

6 In this destruction, several descriptions are mentioned. Burning sensation is attributable to the deficiency of vitamin B12 which occurs because of poor absorption from alimentary canal. Indigestion, weakness, sluggish movement of body parts, tastelessness are attributable to the shift of the distribution of blood from the alimentary canal and musculo skeletal system to heart and brain, thereby he may get the above said symptoms. Emaciation is usually attributable to depletion of adipose tissue which manifests due to poor absorption of nutrients from the alimentary canal.

• This may also be associated with burning sensation, indigestion, weakness, sluggish movement of body parts, tastelessness and emaciation⁷.

• This is manifested due to excess vitiation of Pitta Dosha in the body. This vitiated Pitta may be observed in Koshtha or Shakha.

Samprapti of Kamala



Samprapti Ghataka

- ❖ **Dosha:** Pitta Dosha
- ❖ **Dushya:** Rakta, Mamsa
- ❖ **Srotas:** Raktavaha and Mamsavaha Srotas
- ❖ **Srotodushti Prakara :** Sanga and Vimargagamana
- ❖ **Agni:** Jatharagnimandya and Rakta Dhatvagnimandya
- ❖ **Ama:** Jatharagnimandyaajanya and Raktadhatvagnimandyaajanya
- ❖ **Udbhava Sthana :** Amashaya
- ❖ **Sanchara Sthana :** Sarva Sharira
- ❖ **Vyakta Sthana:** Toak, Mutra, Purisha, Netra, Nakha, Anana
- ❖ **Rogamarga :** Abhyantara, Bahya in the order of manifestation of symptoms

Samanya Lakshana of Kamala

The clinical features of *Kamala* are explained below.

हारिद्रनेत्रः स भृशं हारिद्रत्वङ्मुखाननः।

रक्तपीतशकृन्मूत्र मेकवर्णो हतेन्द्रियः॥3 5॥

दाहाविपाकदौर्बल्यसदनारुचिकर्षितः।

Ca.Ci. – 16/35,36

- ❖ Yellowish discolouration of eyes⁸
- ❖ Yellowish discolouration of skin⁹

- 8 Yellowish discolouration of the eyes is seen in bulbar conjunctiva. This is seen in mucosa covering the sclera of the eye. This is observed after the yellowish discolouration of the ventral surface of the tongue.
- 9 Yellowish discolouration of the skin is seen after the progression of the symptoms from blood, thence to tongue, and later to the sclera of the eye and at last uniform yellowish discolouration in the skin is observed. This usually manifests due to the excess accumulation of bile salts secondary to obstructive jaundice.
- 10 As far as the involvement of buccal mucosa is concerned, abnormal discolouration is first seen in ventral surface of the tongue.
- 11 Reddish discolouration of the urine, eyes or faecal matter can be attributable to the hampered secretion of prothrombin which is continuously secreted by liver cells and lack of this factor can be responsible for bleeding from any part of the body.
- 12 Yellowish discolouration of urine is observed after the elevation of bilirubin in the blood and enhanced conjugated bilirubin in the body. unconjugated bilirubin is not seen in the urine as it is water insoluble.
- 13 Yellowish discolouration of faecal matter refers to the increased presence of bilirubin in the faecal matter. It indicates conjugated bilirubin in the faecal matter and patency of hepato biliary system which implies increased passage of bile from the hepato biliary tree to the duodenum.
- 14 The term refers to loss of function of Jnanendriya and Karmendriya. This is attributable to the nature of the pathological event. For example, when you take the function of Payu (anus), the person will have steatorrhoea (which indicates the abnormal emulsification of fat). When you take the example of Chakshurindriya, the person will have yellowish appearance of the substances as the deposition of bilirubin is seen in the bulbar conjunctiva. It refers to the deposition of bilirubin is seen in conjunctiva covering cornea too, but discolouration is observed in sclera, as cornea is colourless.
- 15 This can be attributable to the peripheral neuropathy and damages to the white matters of the brain and spinal cord. This is occurring as a result of vitamin B12 deficiency. It is being described in the textbook of physiology that vitamin B12 is stored in the liver and this will be stored to last for at least 1 year and may be several years. Thus in case of liver dysfunction along with abnormal bilirubin metabolism, we can observe vitamin B12 deficiency in due course of time.
- 16 Indigestion is the symptom observed in obstructive jaundice. In this type of jaundice, due to the absence of bile secretion into the duodenum, the person will have fat dysmetabolism resulting in steatorrhoea. In due course of time, the person will have disturbances in absorption of fat soluble vitamins.
- 17 Weakness is attributable to carbohydrate and fat dysmetabolism. In the person who is suffering from liver disease, glucose transported from alimentary canal cannot be converted into the glycogen and conversion of glucose to glycogen and vice versa is disturbed. When there is poor fat absorption, that will lead to poor adipose tissue formation. Adipose tissue is considered to be energy reservoir in case of carbohydrate depletion in the body. Because of the above two mechanisms, the person may have weakness.
- 18 As there is nutrition deficiency that is of carbohydrates, fats and proteins, the person suffering from this illness may have sluggish movement of body parts.
- 19 This can be attributable to the next stage of weakness. In this particular stage, due to the poor supply of carbohydrate, there will be depletion of adipose tissue takes place. The area where there will be deposition of adipose tissue is below the skin. Thus, continuous fat depletion from the body can have emaciation in due course of time.

- ❖ Yellowish discolouration of nails
- ❖ Yellowish discolouration of mouth or buccal mucosa¹⁰
- ❖ Reddish¹¹ or yellowish discolouration of urine¹²
- ❖ Reddish or yellowish discolouration of faecal matter¹³
- ❖ Patient's skin resembles the colour of frog observed during rainy season
- ❖ Loss of function of all *Indriya*¹⁴
- ❖ Burning sensation of the body¹⁵
- ❖ Indigestion¹⁶
- ❖ Weakness¹⁷
- ❖ Sluggish movement of body parts¹⁸
- ❖ Tastelessness
- ❖ Emaciation¹⁹

Types of Kamala

Sankhya Samprapti of Kamala 2 types.

1. **Koshthashakhashrita Kamala (Bahupitta Kamala)-** *Svatantra* and *Paratantra Kamala*
2. **Shakhashrita Kamala (Alpapitta Kamala)**

Here, the term *Koshthashakhashrita Kamala* refers to the presence of vitiated *Pitta* in the *Koshtha* as well as

Shakha. That is why, this type of *Kamala* is also termed as *Bahupitta Kamala*. The term *Shakhashrita Kamala* refers to the presence of vitiated *Pitta* in *Shakha* will be there. That is why, this type of *Kamala* is also termed as *Alpapitta Kamala*.

The term *Svatantra Kamala* refers to the manifestation of *Kamala* without the manifestation of *Pandu* who has the vitiation of *Pitta Dosha*. Whereas *Paratantra Kamala* refers to the manifestation of *Kamala* which occurs due to the primary manifestation of *Pandu*.

1. Koshthashakhashrita Kamala

पाण्डुरोगी तु योऽत्यर्थं पित्तलानि निषेवते।

तस्य पित्तमग्मांसं दग्ध्वा रोगाय कल्पते॥34॥

हारिद्रनेत्रः स भृशं हारिद्रत्वदनखाननः।

रक्तपीतशकृन्मूत्रो मेकवर्णो हतेन्द्रियः॥35॥

दाहाविपाकदौर्बल्यसदनारुचिकर्षितः।

कामला बहुपितैषा कोष्ठशाखाया मता॥36॥

Ca.Ci. – 16/34-36

- This is also being termed as *Bahupitta Kamala*.
- *Pitta* vitiating diet is considered to be *Nidana* or etiological factor for this type of *Kamala*.
- First there will be *Pitta* vitiation, followed by affliction of *Rakta* and *Mamsa* involvement.
- **Clinical Features:** The clinical features of *Koshthashakhashrita Kamala*²⁰ are yellowish discolouration of eyes, skin, nails and oral cavity; reddish yellow discolouration of urine and faeces; the skin colour resembles the colour of skin of frog; loss of function of all *Indriya*, burning sensation, indigestion, weakness, sluggish movements of the body parts, tastelessness and emaciation.
- As this is *Bahupitta Kamala* i.e. the disease manifested due to maximum *Pitta* vitiation, this requires aggressive line of treatment.

2. Shakhashrita Kamala Lakshana

तिलपिष्टनिभं यस्तु वर्चः सृजति कामली॥124॥

लेप्यणा रुद्रमार्गे तत् पित्तं कफहरैर्जयेत्।

रुक्षशीतगुरुस्वादुव्यायामैर्वेगनिग्रहैः॥125॥

कफसमूर्च्छितो वायुः स्थानात् पित्तं क्षिपेडली।

हारिद्रनेत्रमूत्रत्वक् श्वेतवचस्तदा नरः॥126॥

भवेत् साटोपविष्टम्भो गुरुणा हृदयेन च।

दौर्बल्याल्पाङ्गिपार्थ्यार्तहिकाश्वासारुचिज्वरैः॥127॥

क्रमेणाल्पेऽनुसज्येतं पित्ते शाखासमाधिते।

Ca.Ci. – 16/124-128

- The causative factors of *Shakhashrita Kamala* are excessive consumption of dry substances, cold substances, food substances heavy to digest, substances having sweet taste, soon after performing exercise; doing suppression of natural urges. All these will lead to the vitiation of *Vata Dosha*, but does the *Murchana* of *Kapha*.
- The consistency of stools will be like paste of *sesamum seeds*²¹.
- **Clinical features:** Clinical features of *Shakhashrita Kamala* include yellowish discolouration of eyes²², urine and skin; whitish discolouration of faecal matter²³; distension of the abdomen; sluggish movement of the intestines; heaviness of the chest; weakness; reduced appetite; pain on either sides of chest; hiccups, breathing discomfort; tastelessness; raised body temperature²⁴.
- The line of treatment should be to follow *Kaphahara* line of treatment first, as there is obstruction of *Kapha* leading to obstruct the movement of *Pitta* in the body.

Kumbha Kamala Lakshana

कालान्तरात् खरीभूता कृच्छ्रा स्यात् कुम्भका मला।

कृष्णपीतशकृन्मूत्रो भृशं शूनश्च मानवः॥137॥

सरक्ताक्षिमुखच्छर्दिविण्मूत्रो यश्च ताम्यति।

दाहारुचितृषानाहतन्द्रामोहसमन्वितः॥138॥

नष्टाग्निसंज्ञः क्षिप्रं हि कामलावान् विपद्यते।

Ca.Ci. – 16/37,38

- *Kumbhakamala* is the next stage of *Koshthashakhashrita Kamala*. The disease enters the chronic stage when the person not receiving any adequate line of treatment in *Koshthashakhashrita Kamala* stage.
- In this stage, the person is presented with blackish or blackish yellow discolouration of urine and faecal matter²⁵.
- Generalised swelling all over the body and abdomen²⁶.

20 The description of *Koshthashakhashrita Kamala* is described earlier in the present chapter.

21 The stool becomes clay coloured due to the lack of stercobilin and other bile pigments. This indicates either intrahepatic or extrahepatic obstruction.

22 This indicates the dark yellowish discolouration of the eyes, urine and skin. This is observed due to the abnormal increase of conjugated bilirubin. As conjugated bilirubin is water soluble, this is observed in the urine. The person will also have dark yellowish tinge of the skin and this is associated with profuse itching secondary to deposition of bile salts in the skin.

23 This is attributable for the absence of bile in the faecal matter which usually gives the yellowish colour to the faecal matter.

24 These symptoms are attributable to the portal hypertension. Due to portal hypertension, the person can develop ascites, which can even stretch the diaphragm, ribs and posteriorly intestines. Thus, because of this pathological mechanism, the person may have distension of the abdomen, heaviness of the chest, reduced appetite and tastelessness, pain on sides of the chest, hiccups and breathlessness.

25 This is attributable to bleeding in the upper alimentary canal secondary to oesophageal varices. It is termed as malena (black tarry stool). At least 100-200 ml of blood in the upper gastrointestinal tract is required to have malena which may persist for several days after cessation of intraalimentary bleeding.

26 This is attributable to the hypoproteinemia resulted as a consequence of hepatic dysfunction. One of the main functions of liver is pro-

- Reddish discolouration of eyes, oral cavity, blood mixed vomiting²⁷; presence of blood in the faecal matter²⁸ and urine.
- Darkness in front of the eyes²⁹.
- Burning sensation in the body; tastelessness; thirst; distension of the abdomen³⁰; drowsiness; altered state of consciousness³¹; loss of appetite³².

Halimaka Lakshana

यदा तु पाण्डोर्वर्णः स्याद्धरितश्यावपीतकः॥132॥

बलोत्साहक्षयस्तन्द्रा मन्त्राग्नित्वं मृदुज्वरः।

स्त्रीप्यपोऽङ्गमर्दश्च श्वासस्तृष्णाऽरुचिर्भ्रमः॥133॥

हलीमकं तदा तस्य विद्यादनिलपित्ततः।

Ca.Ci. – 16/132-134

- *Halimaka* is the disease which is manifested due to the vitiation of *Vata* and *Pitta Dosha*.
- The person will be presented with lack of physical strength; lack of enthusiasm; drowsiness; impaired or loss of appetite; mild rise of body temperature; lack of interest in sexual act or loss of libido³³; bodyache all over the body; breathing discomfort; thirst; tastelessness; giddiness
- The discolouration of skin³⁴ may be either greenish discolouration or blackish discolouration or yellowish discolouration

Panaki Lakshana

प्रोगः। उक्तं हि तन्त्रान्तरे - ' संतापो भिन्नवर्चस्वं बहिरन्ते च पीनता।

पाण्डुता नेत्ररागश्च पानकीलक्षणं मतम्' इति॥132॥133॥-

Chakra, Ca.Ci. – 16/132,133

- *Panaki*³⁵ is another stage of *Kamala* which is mentioned by *Chakrapani* commentary in *Charaka Samhita*.
- The clinical features of *Panaki* are raised body

temperature; improperly formed or loose stools, yellowish discolouration of the body everywhere including urine and stools

- Yellowish white discolouration and reddish discolouration of the eyes³⁶

Upadrava of Kamala³⁷

सरक्ताक्षिमुखच्छर्दिर्विण्मूत्रो यश्च ताम्यति।

दाहारुचितृषानाहतन्द्रामोहसमन्वितः॥138॥

नष्टाग्निसंज्ञः क्षिप्रं हि कामलावान् विपद्यते।

Ca.Ci. – 16/38,39

- Reddish discolouration of eyes
- Reddish discolouration of face or oral cavity
- Blood mixed vomiting
- Blood mixed stools
- Blood mixed urine
- Darkness in front of the eyes
- Burning sensation in the body
- Tastelessness
- Thirst
- Distension of the abdomen
- Drowsiness
- Altered state of consciousness
- Loss of appetite

Approach to the patient of hepatobiliary dysfunction

- The individuals who are suffering from hepatobiliary dysfunction will be presented with yellowish discolouration of skin, sclera, ventral surface of tongue, dark yellowish discolouration of urine and

duction of plasma proteins. In the person who is suffering from chronic liver disease, he may have poor production of proteins, as a result of this, there will be increase in the interstitial compartment secondary to reduced oncotic pressure inside the capillaries.

27 This is attributable as a sequel to the rupture of oesophageal varices.

28 This is indicative of passage of blood from the rectum and lower gastrointestinal bleeding with rapid transit of blood across the intestines.

29 This is attributable to syncopal episodes which may occur as a result of hypovolemic shock. Hypovolemia may occur as a result of hypoproteinemia which leads to the reduction of oncotic pressure which leads to the excess movement of fluid from the capillary to the interstitial spaces.

30 This symptom is attributable to the accumulation of fluid in the peritoneal cavity (ascites) as a consequence of portal hypertension.

31 These symptoms are attributable to acute liver failure which is seen in infective hepatitis or in eclampsia or in Weil's disease. This can be associated with various cranial nerve palsies or delirium and episodic convulsions.

32 This symptom is attributable to the increased secretion in the gastrointestinal lumen as a sequel of portal hypertension.

33 This symptom can be attributable to the erectile dysfunction or feminisation syndrome develops in an individual after chronic alcoholism. This is seen in alcoholic liver disease.

34 This discolouration can be attributable to the excess presence of biliverdin (greenish discolouration) or to the excess presence of bilirubin (yellowish discolouration) or excess accumulation of transferrin or ferritin which is a biproduct of heme destruction (blackish discolouration). Ferritin can be stored in the cells of the liver.

35 As per the clinical features of *Panaki*, any of the infection or inflammation of hepato biliary tree can be considered. Improperly formed or loose stools can be observed in the obstructive pathology of the hepato biliary disease. Yellowish discolouration can be first seen in the blood and in the urine and stools. This indicates the severe progression of the disease. Yellowish white discolouration of the eyes can be observed in increased conjugated bilirubin in the blood.

36 Reddish discolouration of the eyes can be attributable to the hampered production of prothrombin which is continually formed in the liver and it is continuously utilised by the body for blood clotting. Thus, the person may have tendency to have bleeding in any part of the body.

37 The applied aspect of this section is discussed in the context of *Kumbha Kamala*.

faecal matter. This abnormal mechanism is caused by wide variety of etiological factors vary from infections, drug interaction, toxin exposure, ischemia and autoimmune disorders.

Overview of bilirubin metabolism

The breakdown of haeme (a component of haemoglobin) is responsible for the production of bilirubin (an insoluble waste product) and other bile pigments. Bilirubin must be made water soluble to be excreted. This transformation occurs in 5 steps: formation, plasma transport, liver uptake, conjugation and biliary excretion.

1. **Formation:** About 250 to 350 mg of unconjugated bilirubin forms daily; 70-80% derives from breakdown of degenerating (senescent) RBCs and 20-30% (early labelled bilirubin) derives primarily from other haeme proteins in the bone marrow and liver. Haemoglobin is degraded to iron and biliverdin, which is converted to bilirubin.
 2. **Plasma transport:** Unconjugated (indirect) bilirubin is not water soluble; so this will be transported via plasma bound to albumin. This will not be passing through kidneys as the size of bilirubin which is attached to albumin cannot pass through glomerulus. Albumin binding weakens under certain conditions such as acidosis.
 3. **Liver uptake:** The liver takes up bilirubin rapidly but does not take up the attached serum albumin.
 4. **Conjugation:** Unconjugated bilirubin in the liver is conjugated to form mainly bilirubin diglucuronide, or conjugated bilirubin. This reaction is catalysed by the enzyme called glucuronyl transferase. This conjugated bilirubin is considered to be water soluble.
 5. **Bilirubin excretion:** Bilirubin is excreted into bile ductules, interlobar bile ducts, and larger hepatic ducts. This further drains into main hepatic duct, thence to cystic duct and that will be stored in gall bladder. From the gall bladder, the bile is draining into common bile duct and later into second part of duodenum via ampulla of Vater.
- The common clinical manifestations of hepatobiliary dysfunction can be observed in the following categories.
 - Common non specific symptoms such as fatigue, anorexia, nausea and vomiting in severe cases
 - Loose, fatty stools (steatorrhea) can occur when cholestasis prevents sufficient bile from reaching the intestines. They may have risks of deficiencies of fat soluble vitamins such as vitamins A, D, E and K), osteoporosis and bleeding.
 - Fever can be seen in viral or alcoholic hepatitis
 - Jaundice seen in either hemolytic or hepatocellular or obstructive phenomena of hepatobiliary dysfunction

- Right upper quadrant pain due to liver disorders can happen due to passive congestion.
- Erectile dysfunction and feminisation are the phenomena usually the outcomes of effects of alcohol.

Jaundice evaluation

- Severity of the illness is indicated by encephalopathy (e.g. mental status changes) or coagulopathy (easy bleeding, purpura or black tarry stool).
- Portal hypertension may manifest with signs such abdominal collateral vasculature, ascites and splenomegaly).
- Massive upper GI bleeding indicates the oesophageal varices.
- Acute jaundice in the young and healthy individuals suggests acute viral hepatitis
- Acute jaundice after acute drug or toxin exposure in healthy patients is likely due to that substance. Hence drug history or exposure to toxic substances is essential.
- A long history of heavy alcohol usage suggests alcoholic liver disease.
- A personal or family history of recurrent, mild jaundice without findings of hepatobiliary dysfunction suggests a hereditary disorder, usually Gilbert syndrome.
- Gradual onset of jaundice with pruritus, weight loss, clay coloured stools indicates intrahepatic or extrahepatic obstruction.
- Blood tests (bilirubin, aminotransferase, alkaline phosphatase) are essential for the differential diagnosis of jaundice
- Imaging of the abdomen most commonly and sometimes, biopsy or laparoscopy may be of help in the determination of the cause in jaundice.

Key summary of the chapter

- Kamala is the spectrum of the diseases with Pitta Dosha vitiation, with Rakta Dhatu affliction, afflicting Raktavaha Srotas with Sanga and Vimargagamana as the type of Srotodushti involved during the manifestation of the disease.
- Amashaya is the Udbhava Sthana of the disease. Sarvasharira is considered as Sanchara Sthana and Vyakta Sthana of the disease.
- Charaka has mentioned 2 types of Kamala – Koshthashakhashrita (Bahupitta) and Shakhashrita (Alpapitta) Kamala.
- Both types of Kamala are again of two categories –
 1. Svatanttra (primary cause) and
 2. Paratantra (secondary cause).
 Kumbhakamala, Halimaka, Panaki are different stages of Kamala.

Self-Assessment: Questions for Review

Long essays

- Explain the different types of Kamala with Nidana, Samprapti, Lakshana, Upadrava of the disease
- Describe Kamala in detail

Short essays

- Explain Shakhashrita Kamala
- Describe Koshthashakhashrita Kamala
- Write the differences between Koshthashakhashrita and Shakhashrita Kamala

MCQs

1. The Anubandha Dosha in Shakhashrita Kamala is

- a. Vata;
- b. Kapha;
- c. Vata and Kapha;
- d. Vata and Pitta

Answer: (b)

2. The term Bhekavarna refers to the colour of

- a. Frog of rainy season;
- b. Frog of other season;
- c. Cheetah;
- d. Tiger

Answer: (a)

3. The Srotodushti observed in Kamala is

- a. Atipravritti;
- b. Siragranthi;
- c. Sanga only;
- d. Sanga and Vimargagamana

Answer: (d)

4. Whitish coloured stools is observed in

- a. Malaria;
- b. Viral hepatitis;
- c. Carcinoma of head of pancreas;
- d. Hereditary spherocytosis

Answer: (c)

5. Which of the following is an example of pre hepatic jaundice?

- a. Viral hepatitis;
- b. Cirrhosis of liver;
- c. Carcinoma of head of pancreas;
- d. Sickle cell anemia

Answer: (d)

6. Conjugated bilirubin rise seen in which of the following disease?

- a. Gilbert syndrome;
- b. Malaria;

- c. Hereditary spherocytosis;
- d. Hepatitis B

Answer: (d)

7. Tilapishta Nibha Purisha is seen in

- a. Pandu;
- b. Kamala;
- c. Atisara;
- d. Grahani

Answer: (b)

8. Kamala and Halimaka are said to be types of Pandu by

- a. Ksharapani;
- b. Harita;
- c. Chakrapani;
- d. Gangadhar Rai

Answer: (b)

9. The colour of Purisha in Ruddhapatha Kamala is

- a. Pita;
- b. Rakta;
- c. Shveta;
- d. Harita

Answer: (c)

10. Primary Srotas afflicted in the patient of Kamala is

- a. Rasavaha Srotas;
- b. Raktavaha Srotas;
- c. Mamsavaha Srotas;
- d. Mutravaha Srotas

Answer: (b)

Chapter resources

- <https://byjus.com/biology/jaundice-symptoms/>
- Charaka Samhita Chikitsa Sthana 16th chapter - Panduroga Chikitsa Adhyaya
- Sushruta Samhita Uttara Tantra
- Golwalla's medicine for students by Aspi F Golwalla and Sharukh A Golwalla, edited by Milind Y Nadkar, 25th edition, Jaypee Brothers Medical Publishers (P) Ltd, New Delhi, 2017
- The Merck Manual of Diagnosis and Therapy by Robert S Porter and Justin L Kaplan, 19th edition, Merck Sharp & Dohme Corp, New Jersey, 2011
- Chamberlain's symptoms and signs in clinical medicine by Colin Ogilvie and Christopher C Evans, 12th edition, Butterworth-Heinemann Publishers, 1997
- Best & Taylor's physiological basis of medical practice by O P Tandon and Y Tripathi, 13th edition, Wolters Kluwer Health (India) Pvt. Ltd, Gurgaon, 2012
- Textbook of Medical physiology by Arthur C Guyton and John E Hall, 11th edition, Saunders Elsevier publications, Philadelphia, 2006
- Physical diagnosis - a textbook of symptoms and physical signs by Rustom Jal Vakil and Aspi F Golwalla, 13th edition, MPP Pvt. Ltd, Mumbai, 2010.